Seminar 11 – GR312CD – 16.12.2021.

- descompunerea în valori singulare
- ecuații diferențiale liniare de ord I

Descompunerea în valori singulare (SVD)

$A \in M_{m,n}(\mathbb{R}) \Rightarrow \exists\, U \in M_m(\mathbb{R})$; $V \in M_n(\mathbb{R})$ ortogonale,

$\Rightarrow \exists\, \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ a.î. $U \cdot U^t = I_m$; $V \cdot V^t = I_n$.

$$\underset{m\times m}{U^t} \underset{m\times n}{A} \underset{n\times n}{V} = \left(\begin{array}{c|c} \begin{matrix}\sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r\end{matrix} & 0 \\ \hline 0 & 0 \end{array}\right)\Big\}_{m-r} = \underset{m\times n}{\Sigma} \Rightarrow \boxed{A = U\Sigma V^t}$$

(r col; $n-r$)

$\sigma_1, \dots, \sigma_r$ valori singulare.

Algoritm pt determinarea SVD. $\Rightarrow \boxed{A = U\Sigma V^t}$

(1) Se ortodiagonalizează matr. sim $B = \underset{n\times m}{A^t}\underset{m\times n}{A}$. $\Rightarrow \underset{=}{V}: n\times n$.

$B^t = (A^tA)^t = A^t(A^t)^t = A^tA = B$

$B = QD\cdot Q^t$; $Q = [v_1 | \dots | v_n] = V$

($\downarrow$ ortogonală)

$v_1, \dots, v_n$ ortonormați.

$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$; $\lambda_{r+1} = \dots = \lambda_n = 0$.

$r = \operatorname{rg} A = \operatorname{rg} A^t$.

(2) $\boxed{\sigma_i = \sqrt{\lambda_i}\ ;\ i = \overline{1,r}} \Rightarrow \Sigma = \left(\begin{array}{c|c} \operatorname{diag}(\sigma_1, \dots, \sigma_r) & 0 \\ \hline 0 & 0 \end{array}\right)$.

$V = (v_1 | \dots | v_n)$.

(3) $U = (u_1 | \dots | u_r | u_{r+1} | \dots | u_m)$.

$\boxed{u_i = \frac{1}{\sigma_i} A v_i}$; $i = \overline{1,r} \Rightarrow \{u_1, \dots, u_r\}$ ortonormați $\xrightarrow{\text{completăm}} \{u_1, \dots, u_m\}$ bază ortonormată pt $\mathbb{R}^m$.

Ex 1 $A=\begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}$? valori singulare și descr SVD.

Γ $\underset{3\times2}{A}=\underset{3\times3}{U}\underset{3\times2}{\Sigma}\underset{2\times2}{V^t}$; $\operatorname{rg}A=2=r$.

① $B=A^tA=\begin{pmatrix}1&1&0\\1&0&1\end{pmatrix}\cdot\begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}=\begin{pmatrix}2&1\\1&2\end{pmatrix}$.

$P_B(\lambda)=\det(B-\lambda I_2)=\begin{vmatrix}2-\lambda&1\\1&2-\lambda\end{vmatrix}=(2-\lambda)^2-1^2=(2-\lambda-1)(2-\lambda+1)$

$=(1-\lambda)(3-\lambda)$.

$\lambda_1=3 \Rightarrow \sigma_1=\sqrt{3}$.
$\lambda_2=1 \Rightarrow \sigma_2=1$ } valori singulare.

② $\Sigma=\begin{pmatrix}\sigma_1&0\\0&\sigma_2\\0&0\end{pmatrix}=\begin{pmatrix}\sqrt{3}&0\\0&1\\0&0\end{pmatrix}$.

$V=(v_1|v_2)$. $v_1\perp v_2$; $\|v_1\|=\|v_2\|=1$.

$\boxed{\lambda_1=3}$ $\Rightarrow ?\ v=\begin{pmatrix}x\\y\end{pmatrix}$ i. $Bv=\lambda_1 v \Rightarrow \begin{pmatrix}2&1\\1&2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=3\begin{pmatrix}x\\y\end{pmatrix}\Rightarrow$

$\Rightarrow 2x+y=3x \Rightarrow y=x \Rightarrow v=\begin{pmatrix}x\\x\end{pmatrix}=x\begin{pmatrix}1\\1\end{pmatrix}$.

$v_1=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$.

$\boxed{\lambda_2=1}$ $\Rightarrow ?\ v=\begin{pmatrix}x\\y\end{pmatrix}$ i. $Bv=\lambda_2 v \Rightarrow \begin{pmatrix}2&1\\1&2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=1\cdot\begin{pmatrix}x\\y\end{pmatrix}\Rightarrow$

$\Rightarrow 2x+y=x \Rightarrow y=-x$. $\Rightarrow v=\begin{pmatrix}x\\-x\end{pmatrix}=x\begin{pmatrix}1\\-1\end{pmatrix}\Rightarrow$

$\Rightarrow v_2=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}=\begin{pmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{pmatrix}$

Obs: B simetrică $\left.\begin{matrix}\\ \lambda_1\neq\lambda_2\end{matrix}\right]\Rightarrow v_1\perp v_2$.

$V=(v_1|v_2)=\begin{pmatrix}1/\sqrt{2}&1/\sqrt{2}\\1/\sqrt{2}&-1/\sqrt{2}\end{pmatrix}$.

5 ③ ? $U=(u_1|u_2|u_3)$; $r=2=\operatorname{rg}A$.

$u_i=\frac{1}{\sigma_i}Av_i$; $i=\overline{1,2}$.

$u_1=\frac{1}{\sqrt{3}}\cdot\begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}\cdot\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}=\frac{1}{\sqrt{3}}\cdot\begin{pmatrix}2/\sqrt{2}\\1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}=\begin{pmatrix}2/\sqrt{6}\\1/\sqrt{6}\\1/\sqrt{6}\end{pmatrix}$.

$\|u_1\|=1$.

$u_2=\frac{1}{\sigma_2}Av_2=\frac{1}{1}\cdot\begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{pmatrix}=\begin{pmatrix}0\\1/\sqrt{2}\\-1/\sqrt{2}\end{pmatrix}$.

$\langle u_1,u_2\rangle=0$

$\{u_1,u_2\}$ $\xrightarrow{\text{completăm}}$ $\{u_1,u_2,u_3\}$ bază ortonormată $\text{în } \mathbb{R}^3$

? w i. $w\perp u_1 \Leftrightarrow \langle w,u_1\rangle=0$

$w\perp u_2 \Leftrightarrow \langle w,u_2\rangle=0$.

$w=\begin{pmatrix}a\\b\\c\end{pmatrix}\Rightarrow \begin{matrix}2a+b+c=0\\ b-c=0\Rightarrow b=c\end{matrix}\Rightarrow 2a=-2c$. $a=-c$.

$\Rightarrow w=\begin{pmatrix}-c\\c\\c\end{pmatrix}=c\begin{pmatrix}-1\\1\\1\end{pmatrix}\Rightarrow u_3=\frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}$.

$U=(u_1|u_2|u_3)=\begin{pmatrix}2/\sqrt{6}&0&-1/\sqrt{3}\\1/\sqrt{6}&1/\sqrt{2}&1/\sqrt{3}\\1/\sqrt{6}&-1/\sqrt{2}&1/\sqrt{3}\end{pmatrix}\Rightarrow \boxed{A=U\Sigma V^t}$.

Ex 2 $A=\begin{pmatrix}1&1&0\\1&0&1\end{pmatrix}$. ? SVD.

$A=\underset{2\times3}{U}\,\underset{2\times2}{\Sigma}\,\underset{2\times3}{V}\,\underset{3\times3}{}^t$.

① $B=A^tA=\begin{pmatrix}1&1\\1&0\\0&1\end{pmatrix}\begin{pmatrix}1&1&0\\1&0&1\end{pmatrix}=\begin{pmatrix}2&1&1\\1&1&0\\1&0&1\end{pmatrix}$

$B = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. $P_B(\lambda) = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} =$

$= (2-\lambda)(1-\lambda)^2 - (1-\lambda) - (1-\lambda) = (1-\lambda)\left[(2-\lambda)(1-\lambda) - 2\right] =$

$= (1-\lambda)\left[\lambda^2 - 3\lambda\right] = \underline{(1-\lambda)\lambda(\lambda-3)}.$

$\begin{cases} \lambda_1 = 3 \Rightarrow \sigma_1 = \sqrt{3} \\ \lambda_2 = 1 \Rightarrow \sigma_2 = 1 \\ \lambda_3 = 0 \end{cases}$ $\underline{r = \operatorname{rg} A = 2}$.

$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} : 2 \times 3.$

$\Sigma = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$

$\boxed{\lambda_1 = 3} \Rightarrow ? \; v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \neq 0. \; Bv = \lambda_1 \cdot v \Leftrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 3 \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$

$\Rightarrow 2x + y + z = 3x.$

$x + y = 3y \Rightarrow x = +2y$
$x + z = 3z \Rightarrow x = +2z$ $\Big| \Rightarrow \boxed{y = z}. \Rightarrow v = \begin{pmatrix} +2y \\ y \\ y \end{pmatrix} = y \begin{pmatrix} +2 \\ 1 \\ 1 \end{pmatrix}.$

$v_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} +2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}.$

$\underline{\lambda_2 = 1}. \Rightarrow ? \; v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \neq 0. \; Bv = \lambda_1 v \Leftrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 1 \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$

$x + y = y \cdot \Rightarrow \boxed{x = 0.}$
$x + z = z$
$2x + y + z = x.$ $\Big\} \Rightarrow y + z = 0. \Rightarrow \boxed{z = -y}.$

$v = \begin{pmatrix} 0 \\ y \\ -y \end{pmatrix} = y \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \cdot \Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}.$

$\underline{\lambda_3 = 0}. \Rightarrow ? \; v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \neq 0. \begin{cases} 2x + y + z = 0 \\ x + y = 0 \\ x + z = 0 \end{cases} \Rightarrow \begin{matrix} y = z \\ x = -z \end{matrix} \Rightarrow v = \begin{pmatrix} -z \\ z \\ z \end{pmatrix} = z \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$

$\Rightarrow v_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

$$V = (v_1 | v_2 | v_3) = \begin{pmatrix} +2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}.$$

③ $U = (u_1 | u_2) : 2 \times 2.$

$$u_i = \frac{1}{\sigma_i} A v_i \; ; \; i = \overline{1,2}.$$

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} +2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix} = \frac{1}{\sqrt{3}} \cdot \begin{pmatrix} 3/\sqrt{6} \\ +3/\sqrt{6} \end{pmatrix} = \begin{pmatrix} +\frac{3}{3\sqrt{2}} \\ +\frac{3}{3\sqrt{2}} \end{pmatrix}$$

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{1} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}.$$

$$u_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \Rightarrow U = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}.$$

$$A = U \Sigma V^t \Rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}}_{U} \underbrace{\begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}}_{V^t}$$

Ecuații diferențiale

• necunoscuta e o fct → ? $f(x)$
• ec implică derivate ale fct nec

? f ĩ.
$f''+f'+f=e^x$, $\forall x\in\mathbb{R}$.
$f''(x)+f'(x)+f(x)-e^x=0$, $\forall x\in\mathbb{R}$.

fct nec = fct de o sg. var.
→ ec. diferențiale ordinare EDO

fct. nec = fct de mai multe var. $f(x_1,\dots,x_n)$
→ $\frac{\partial f}{\partial x_i}$; $\frac{\partial^2 f}{\partial x_i^2}$ ⇒ ec. cu derivate parțiale EDP.

• **Def.** O ec. dif de ordin n
F cunoscută

$F(x, y, y'(\cdot), \dots, y^{(n)}(\cdot)) = 0$; $\forall x\in I$ (*)
var indp → $y(x)$ fct nec.

exp: $F(x,y,y',y'') = y''+y'+y-e^x$

• **Def.** O sol pt ec dif (*) e o fct φ de cls $C^n(I)$.
ĩ. $F(x, \varphi(x), \varphi'(x), \dots, \varphi^{(n)}(x)) = 0$; $\forall x$.

• **Exp.** $y''=2y'-y$. Verif că $y_1=e^x$; $y_2=xe^x$ sol

$y_1=e^x \Rightarrow y_1'=e^x=y_2'' \Rightarrow 2y_1'-y_1=2e^x-e^x=e^x=y_1''$.

Ec dif. de ordin I

$F(x,y,y')=0$. Putem pp că putem explicita y' fct de x,y.

$$\boxed{y'=f(x,y).}$$

a) Ec. cu var. separabile $\boxed{y'=f(x)\cdot g(y)}$ f,g cunoscute. $y(\cdot)$ nec.

$\underset{g(y)\neq 0}{\Rightarrow}$ $\frac{1}{g(y)}\cdot y'(x)=f(x) \Rightarrow \int \frac{1}{g(y)}y'(x)\,dx=\int f(x)dx+C$.
⇒ sol implicită în y.

Ex 1. $y'=\underbrace{\sin x}_{f(x)}\cdot\underbrace{\cos^2 y}_{g(y)}$ ec. cu var sep ⇒ $\frac{y'}{\cos^2 y}=\sin x$ / $\int dx$.

$\Rightarrow \int \frac{y'(x)}{\cos^2 y(x)}dx=\int \sin x\,dx \Rightarrow \boxed{\operatorname{tg} y=-\cos x+C}$

$y\neq(2k+1)\cdot\frac{\pi}{2}$; $k\in\mathbb{Z}$.

Ec dif liniare de ord I:

$$y' = f(x)\cdot y + g(x)$$

Exp: $y' = \underbrace{2x}\cdot y + \underbrace{e^x}$ lin; $y'\cdot y = 3\ln|x|$; $y' = e^x y - 2$.
nu e lin — lin

$y' = y^2\cdot x + y\cdot x^2$ nu e lin

I) Etapa omogenă $y' = f(x)\cdot y \Rightarrow \int \frac{y'}{y} = \int f(x) \rightarrow \ln|y| = \int f(x)dx + C$.

$$y_o(x) = k\cdot e^{\int f(x)dx}$$

$|y| = e^{\int f(x)dx}\cdot e^C$.

II) Etapa neomogenă ⇒ Met. variației constantelor Lagrange

$y(x) = k(x)\cdot e^{\int f(x)dx}$ sol pt $y' = f(x)\cdot y + g(x)$.

$$y' = k'(x)\,e^{\int f(x)dx} + k(x)\cdot e^{\int f(x)dx}\cdot f(x) = (k' + k\cdot f)\,e^{\int f(x)dx}$$

$$y' = f(x)\cdot k(x)e^{\int f(x)dx} + g(x).$$

$$\Rightarrow k'\cdot e^{\int f(x)dx} + k\cdot f\cdot e^{\int f(x)dx} = f\cdot k\cdot e^{\int f(x)dx} + g(x)$$

$$k'(x) = e^{-\int f(x)dx}\cdot g(x).$$

$$K(x) = \int g(x)\,e^{-\int f(x)dx} + C.$$

$$y(x) = y_o(x) + y_p(x).$$

$$y(x) = e^{\int f(x)dx}\left[C + \int g(x)e^{-\int f(x)dx}dx\right].$$

Exp: $xy' - 2y = 2x^4$

$xy' = 2y + 2x^4 \Rightarrow y' = \underbrace{\frac{2}{x}}_{f(x)}y + \underbrace{2x^3}_{g(x)}$. $= y\cdot f(x) + g(x)$.

$f(x) = \frac{2}{x}$; $g(x) = 2x^3$; $x \neq 0$.

$$y(x) = e^{\int \frac{2}{x}dx}\left[C + \int 2x^3\cdot e^{-\int \frac{2}{x}dx}dx\right] \Rightarrow$$

$\int \frac{2}{x}dx = 2\ln|x| \Rightarrow \int 2x^3 e^{-\int \frac{2}{x}dx}dx = \int 2x^3\cdot \underbrace{e^{-2\ln|x|}}_{x^{-2}}dx = \int 2x$

$= x^2$.

$$y(x) = x^2\left[C + x^2\right] = x^2\cdot C + x^4.$$

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Ex2: (a) $xy' - y + x = 0$; (b)! $y' + \frac{2y}{x^2-1} = 2x+2$ $y(0) = -3$; (c) $y' = y\cdot ctgx + 2x\sin x$